

Machine Learning for Business

GTSC2143

Lecture 4 Regression I

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AY2025-2026 Semester 1

Case: Predicting House Prices

Predicting house prices

How much is my house worth?



I want to list
my house for
sale

Predicting house prices

How much is my house worth?



Predicting house prices

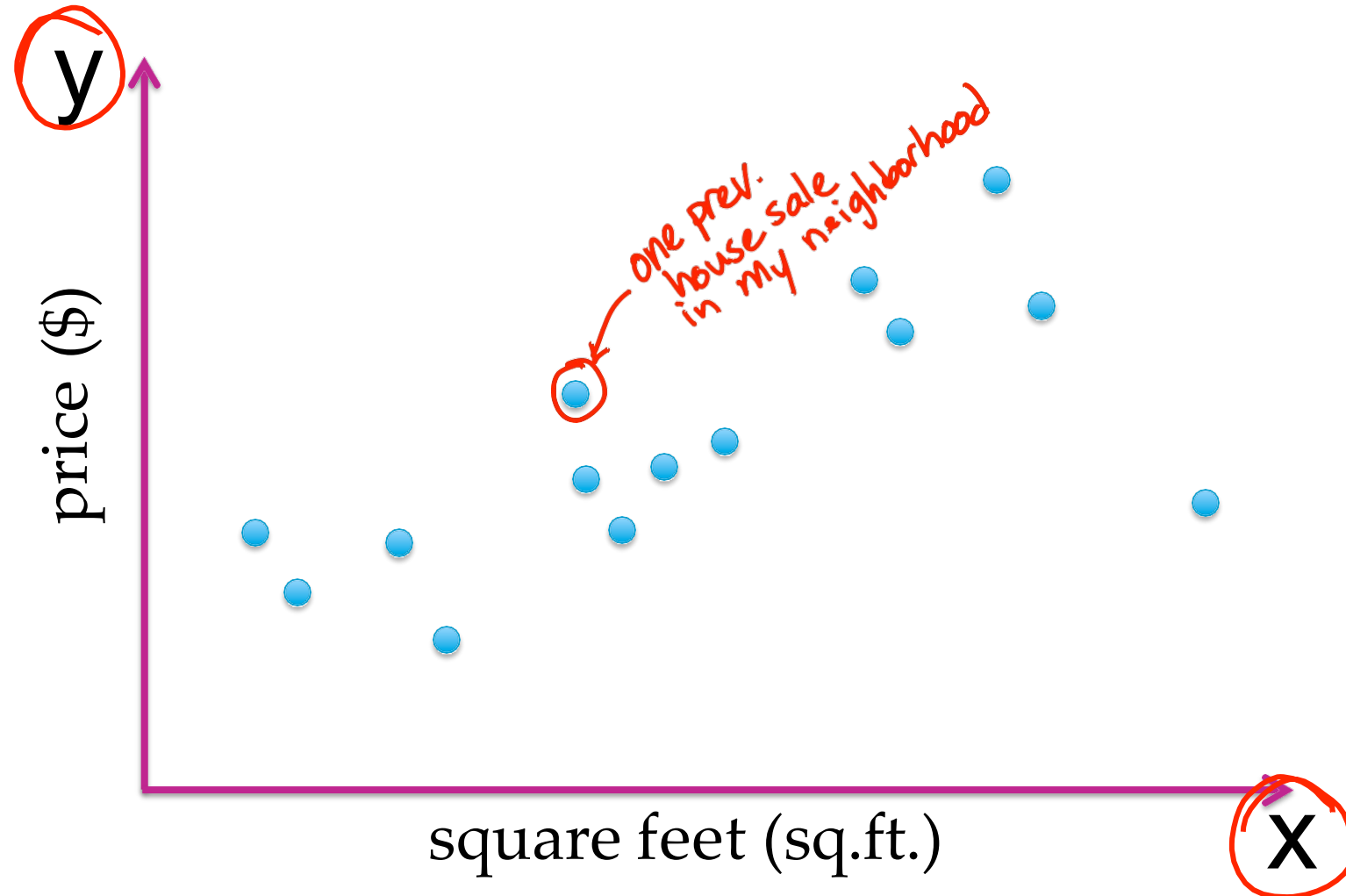
Look at recent sales in my neighborhood

- How much did they sell for?



Predicting house prices

Plot recent house sales (Past 2 years)



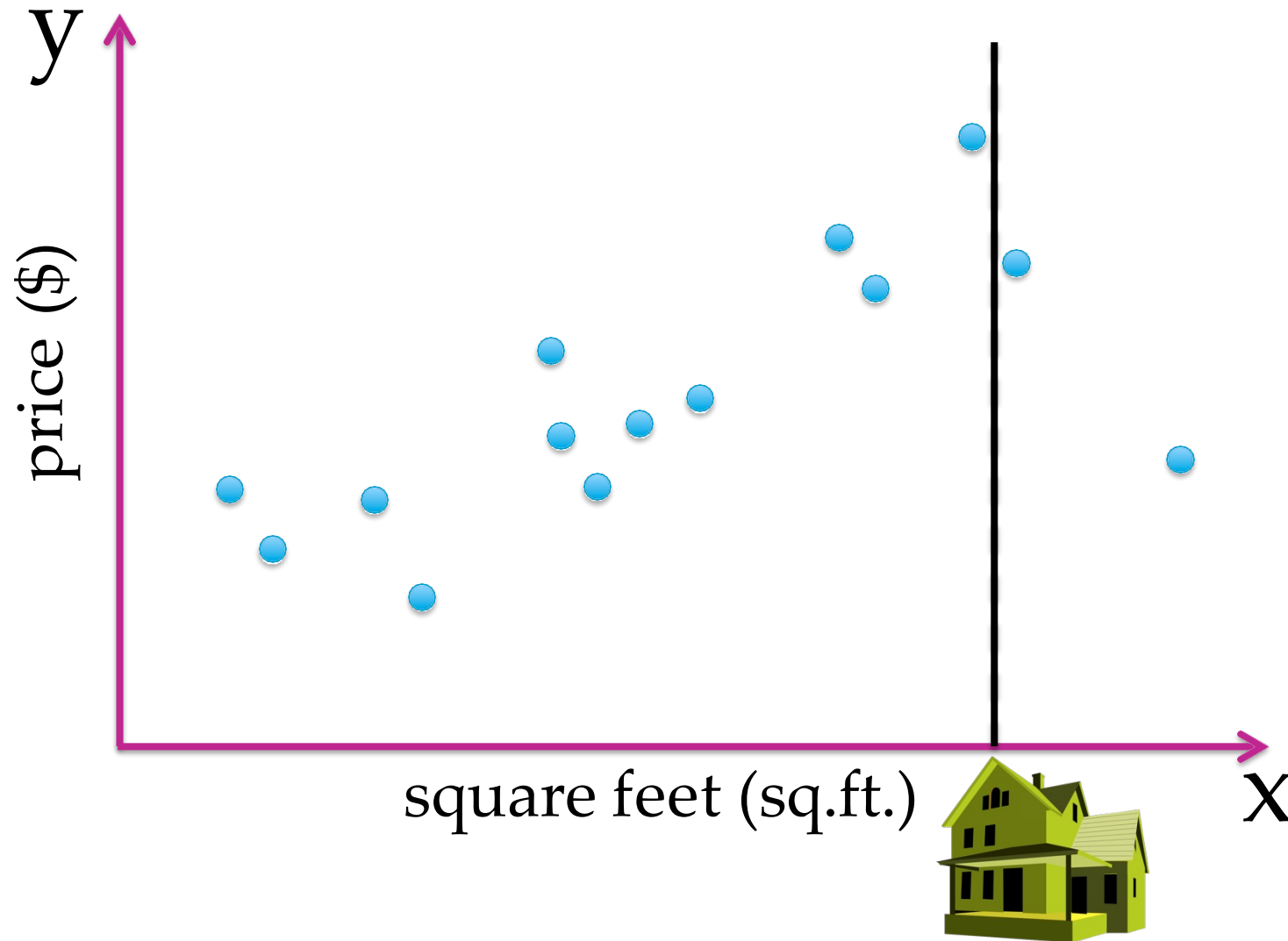
Terminology:

x – feature,
covariate, or
predictor

y – observation or
response

Predicting house prices

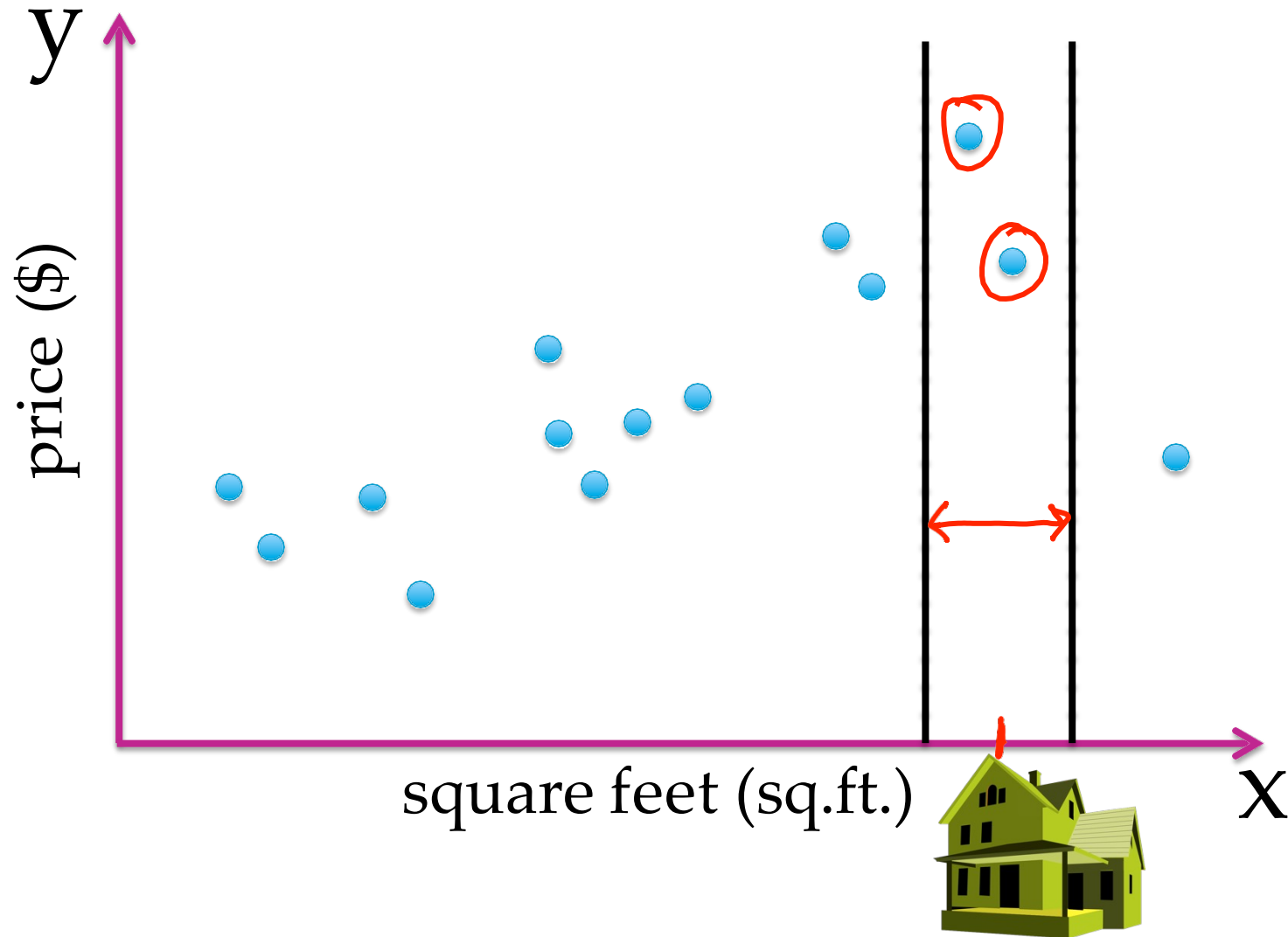
Predict your house by similar houses



- No house sold recently had *exactly* the same sq.ft.

Predicting house prices

Predict your house by similar houses

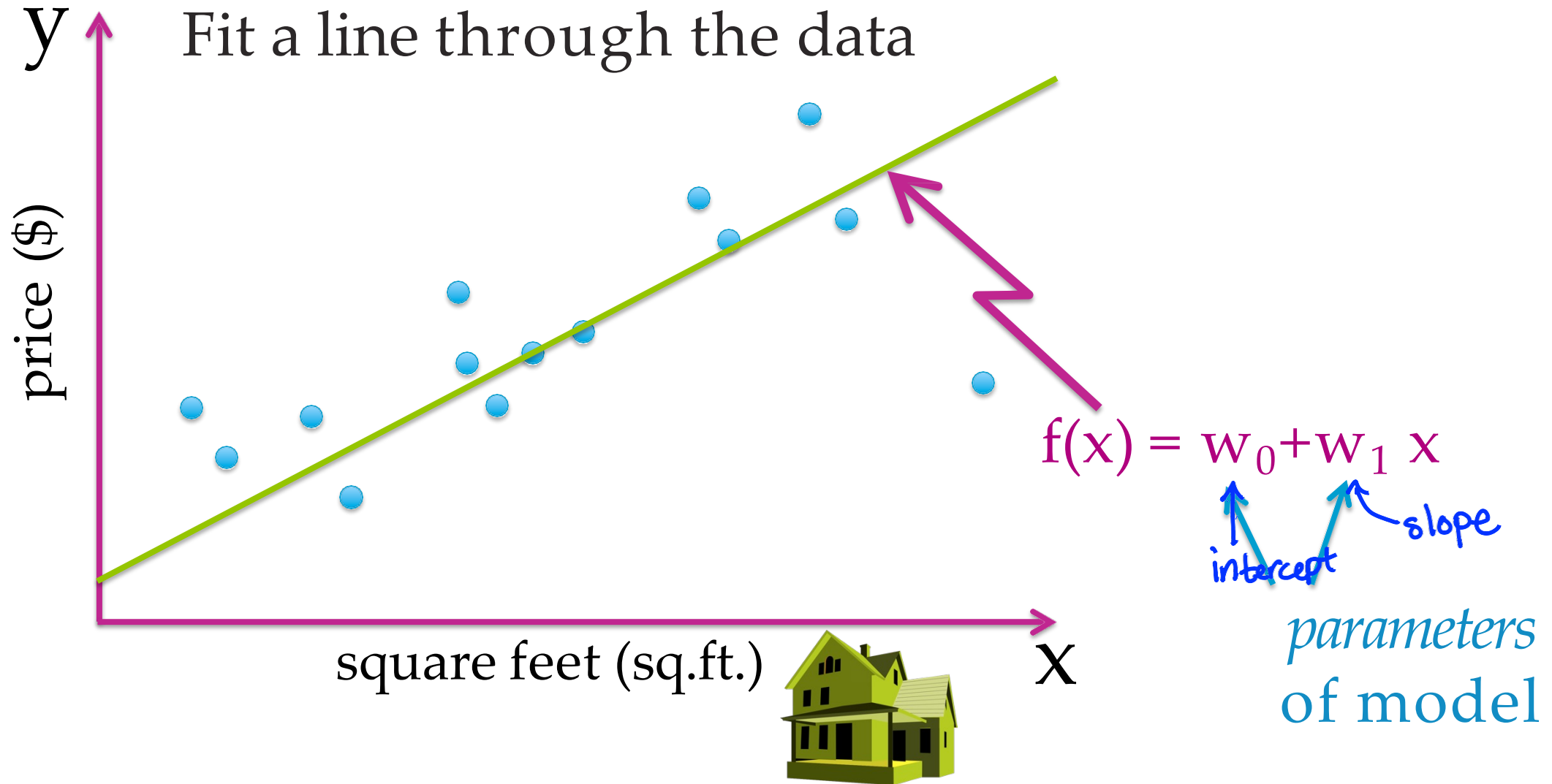


- Look at average price in range
- Still only 2 houses!
- Throwing out info from all other sales

Linear Regression

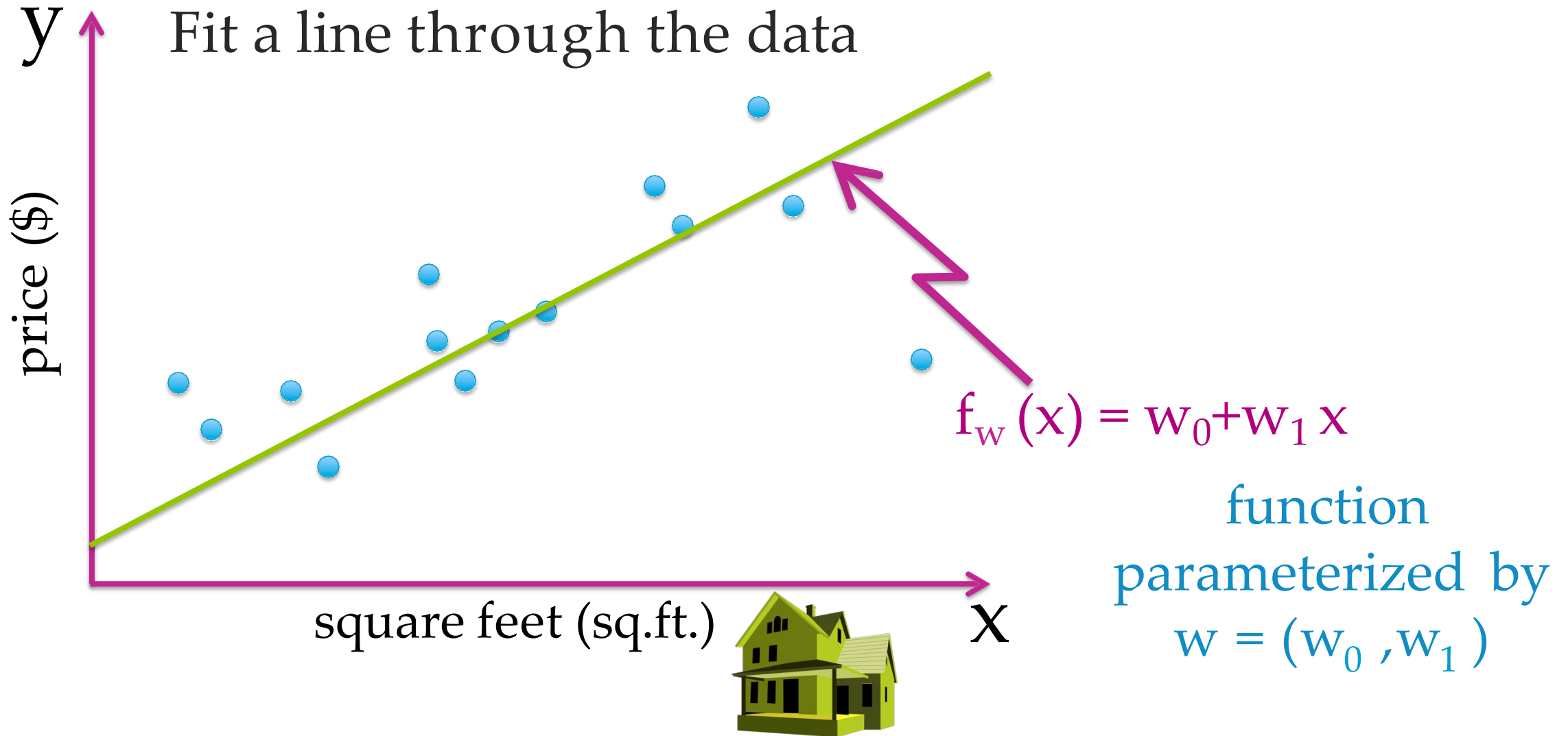
Predicting house prices

Use a linear regression model



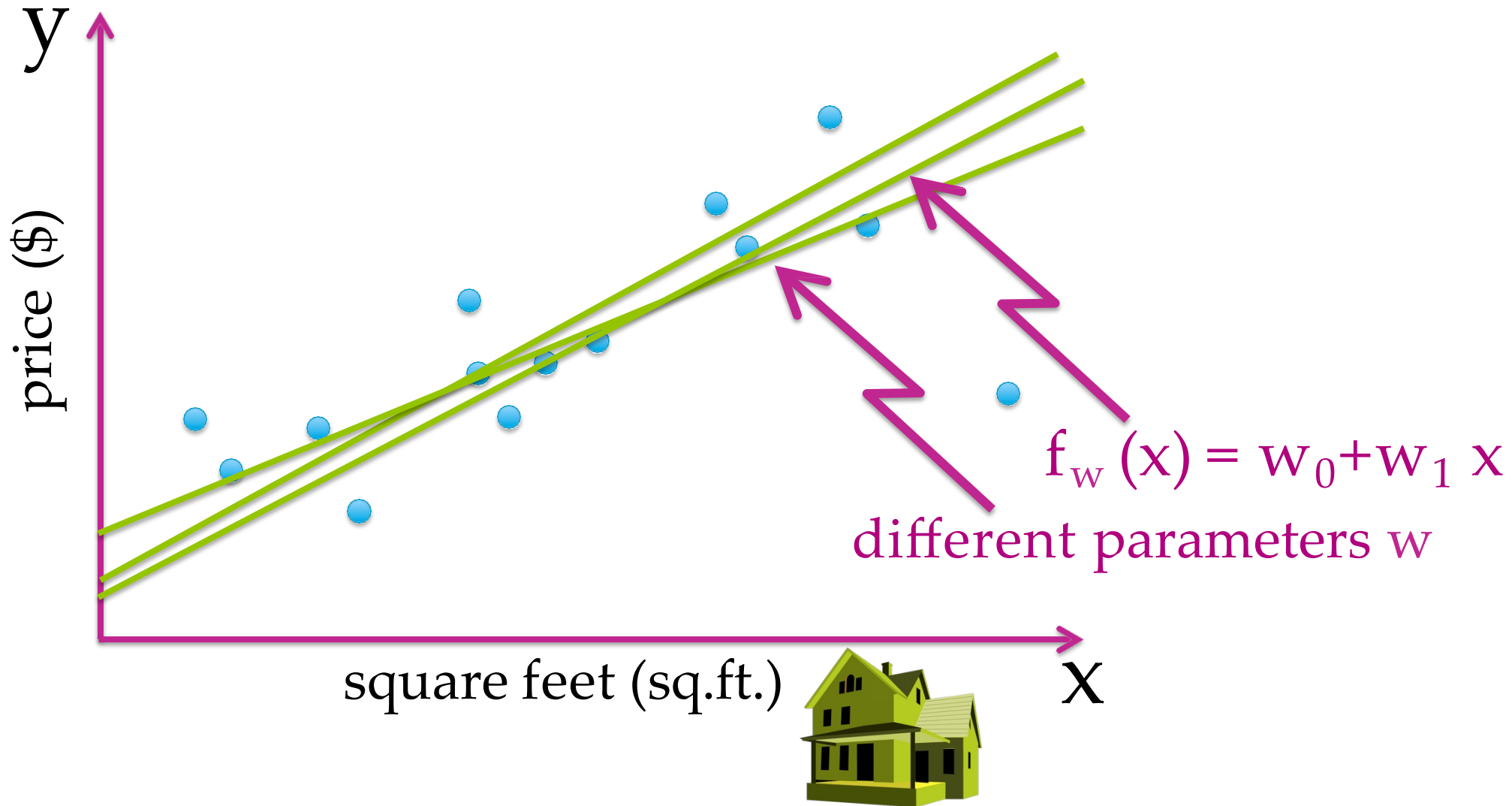
Predicting house prices

Use a linear regression model



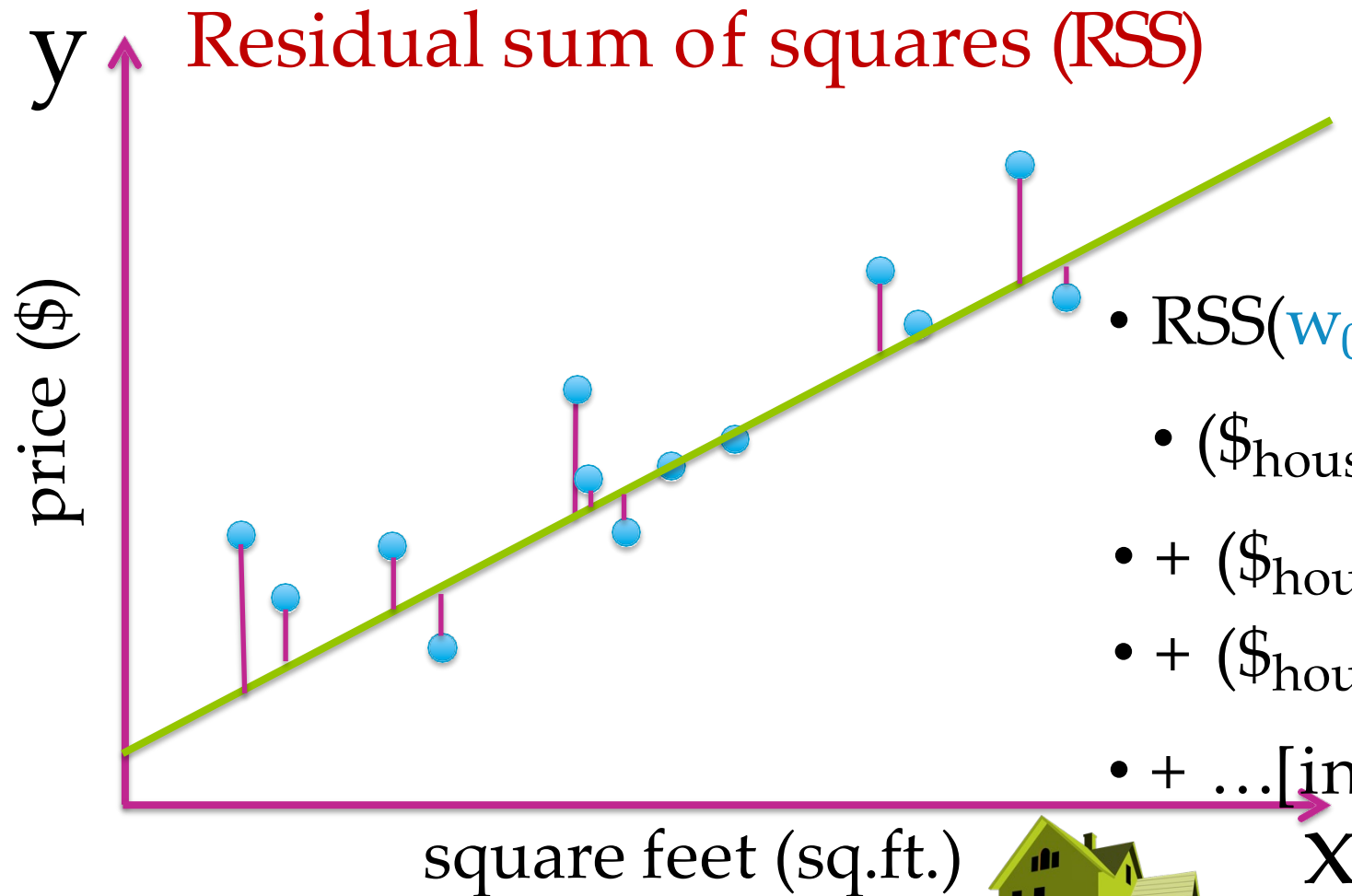
Predicting house prices

But which line should we use?



Predicting house prices

“Cost” of using a given line



- $RSS(w_0, w_1) =$

- $(\$_{\text{house 1}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 1}}])^2$

- $+ (\$_{\text{house 2}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 2}}])^2$

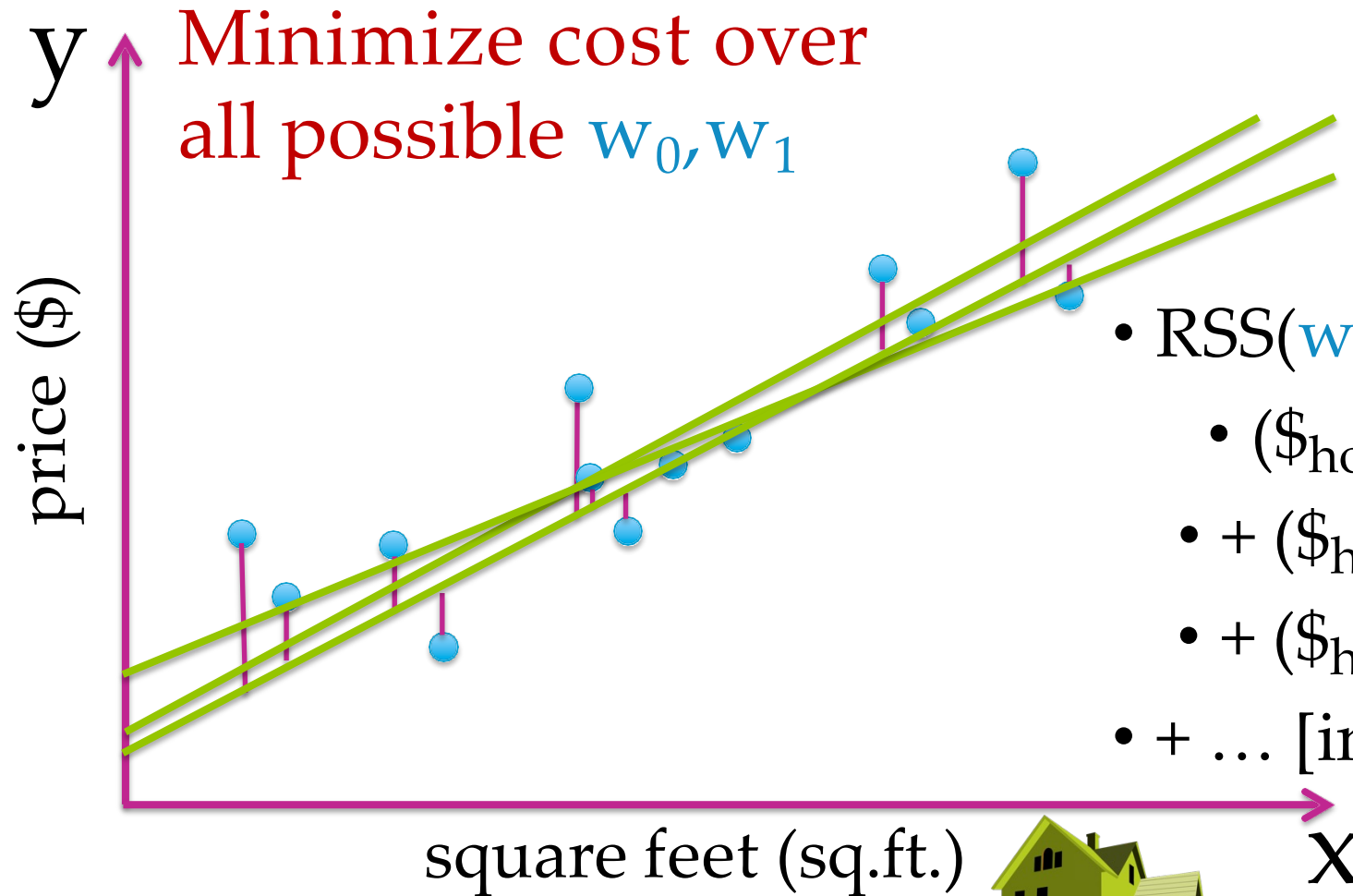
- $+ (\$_{\text{house 3}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 3}}])^2$

- $+ \dots [\text{include all houses}]$



Predicting house prices

Find “best” line



- $RSS(w_0, w_1) =$

- $(\$_{\text{house 1}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 1}}])^2$

- $+ (\$_{\text{house 2}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 2}}])^2$

- $+ (\$_{\text{house 3}} - [w_0 + w_1 \text{sq.ft.}_{\text{house 3}}])^2$

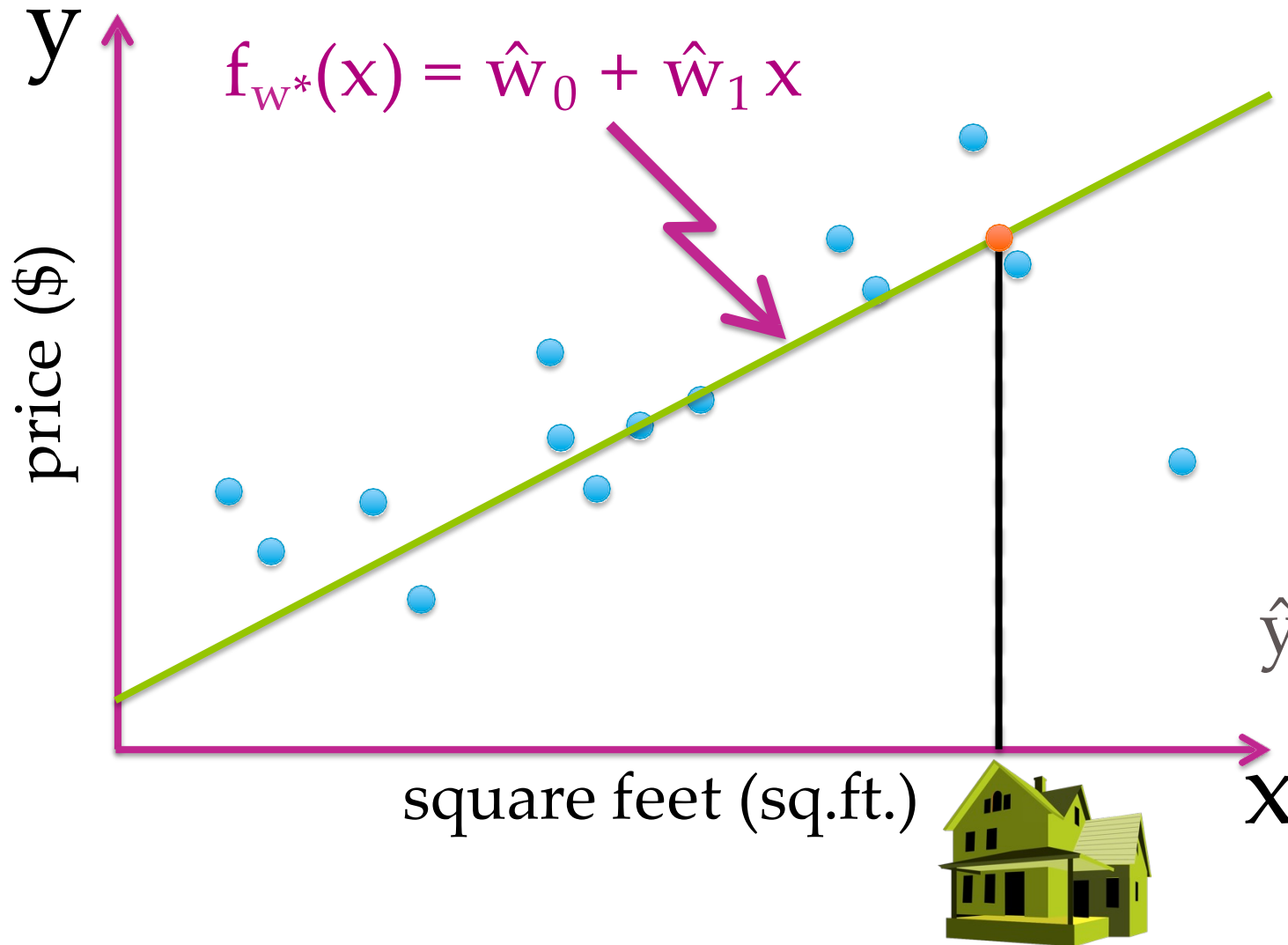
- $+ \dots$ [include all houses]



$\hat{w}$

Predicting house prices

Predicting your house price



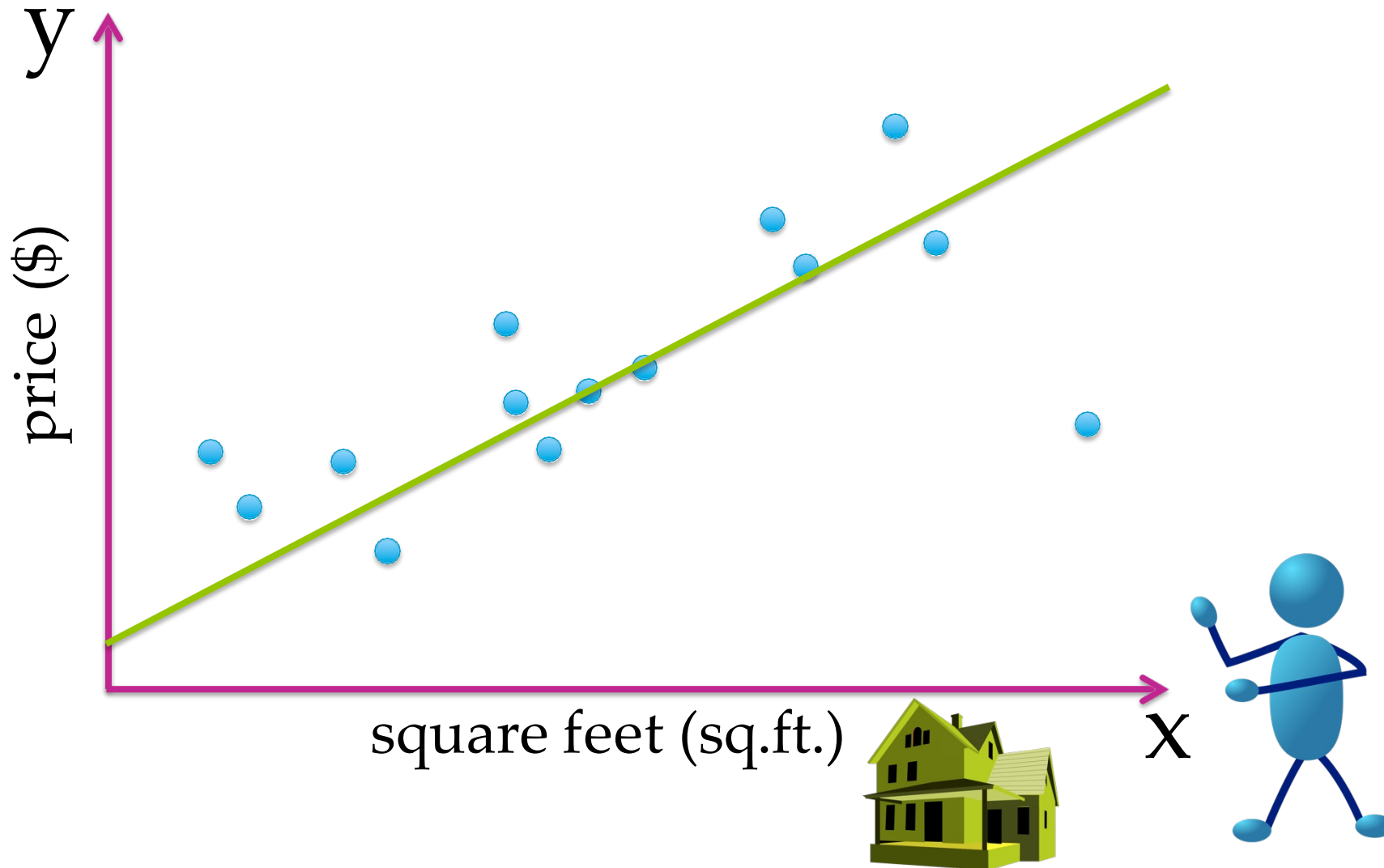
Best guess of your
house price:

$$\hat{y} = \hat{w}_0 + \hat{w}_1 \text{sq.ft.}_{\text{your house}}$$

Adding Higher Order Effects

Predicting house prices

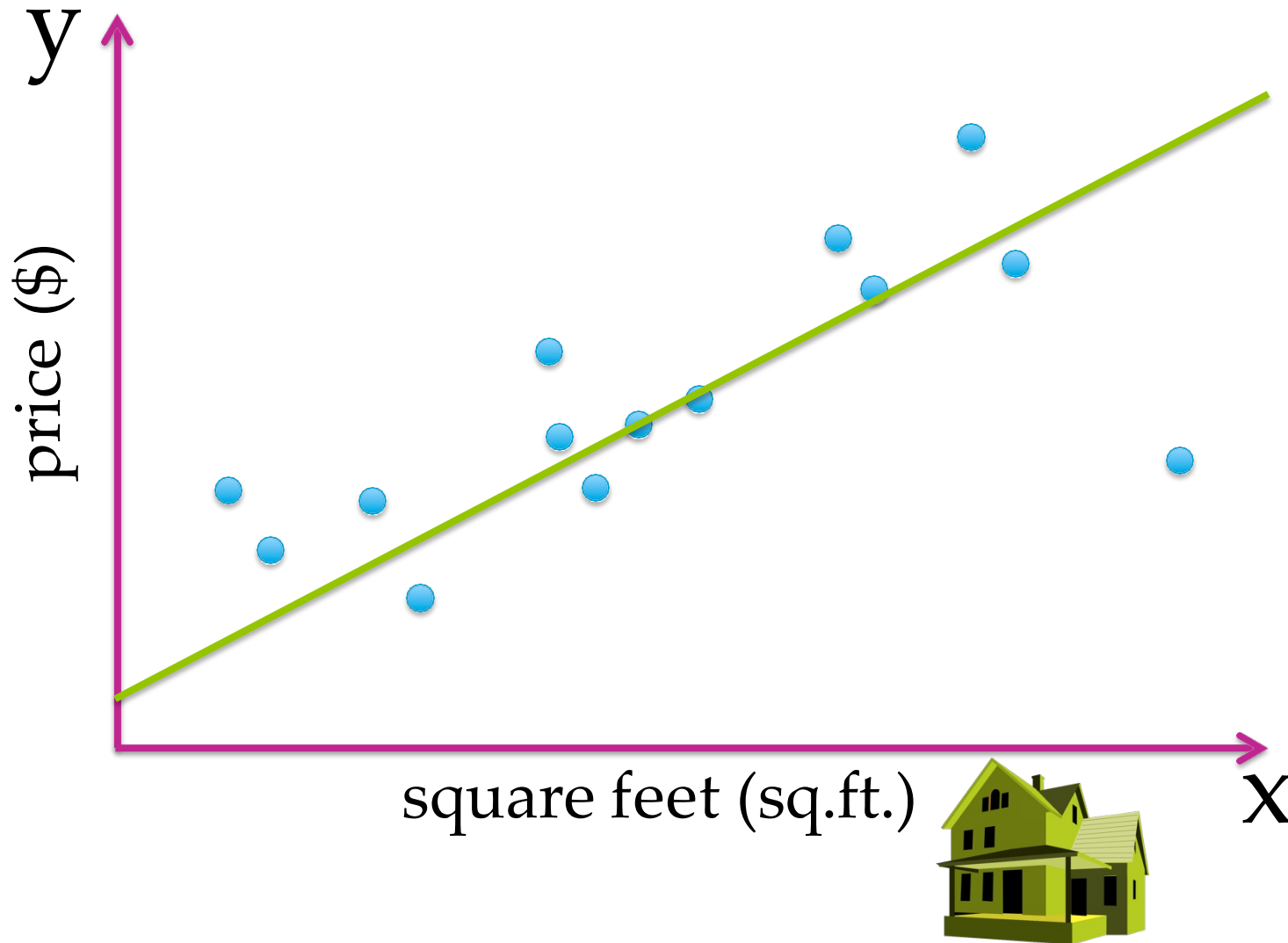
Fit data with a line or ... ?



You show
your friend
your analysis

Predicting house prices

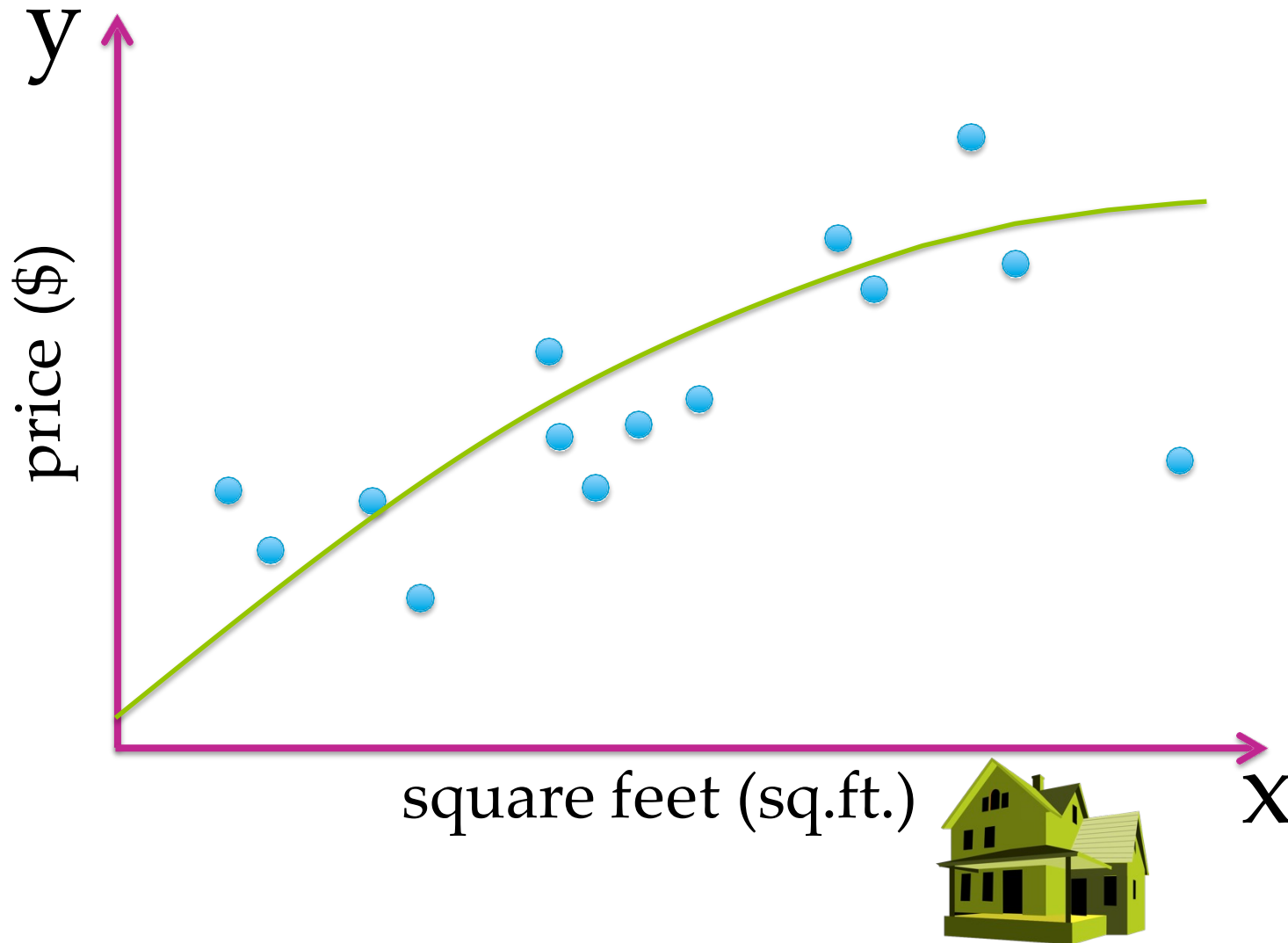
Fit data with a line or ... ?



Dude, it's not
a linear
relationship!

Predicting house prices

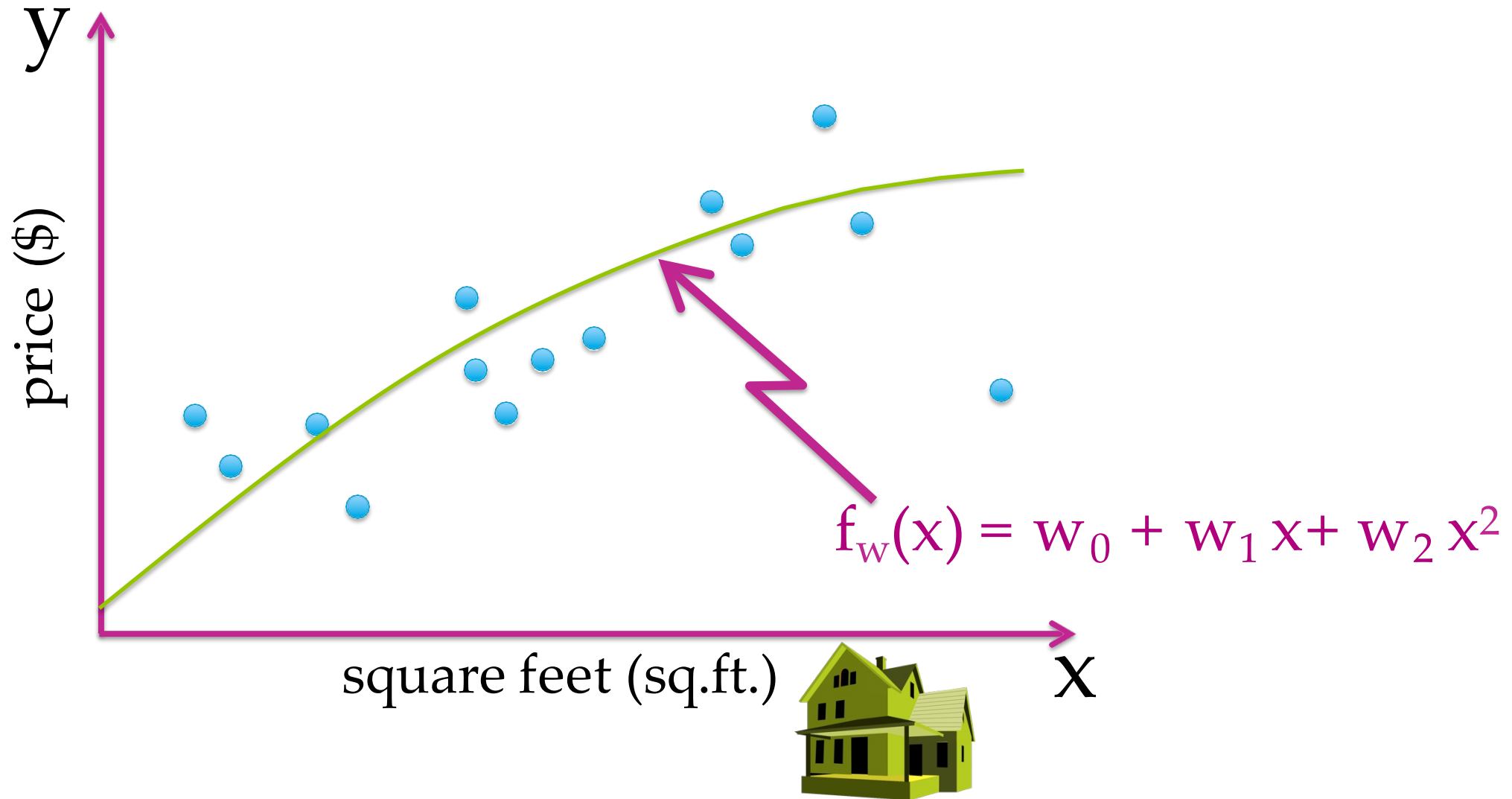
What about a quadratic function?



Dude, it's not
a linear
relationship!

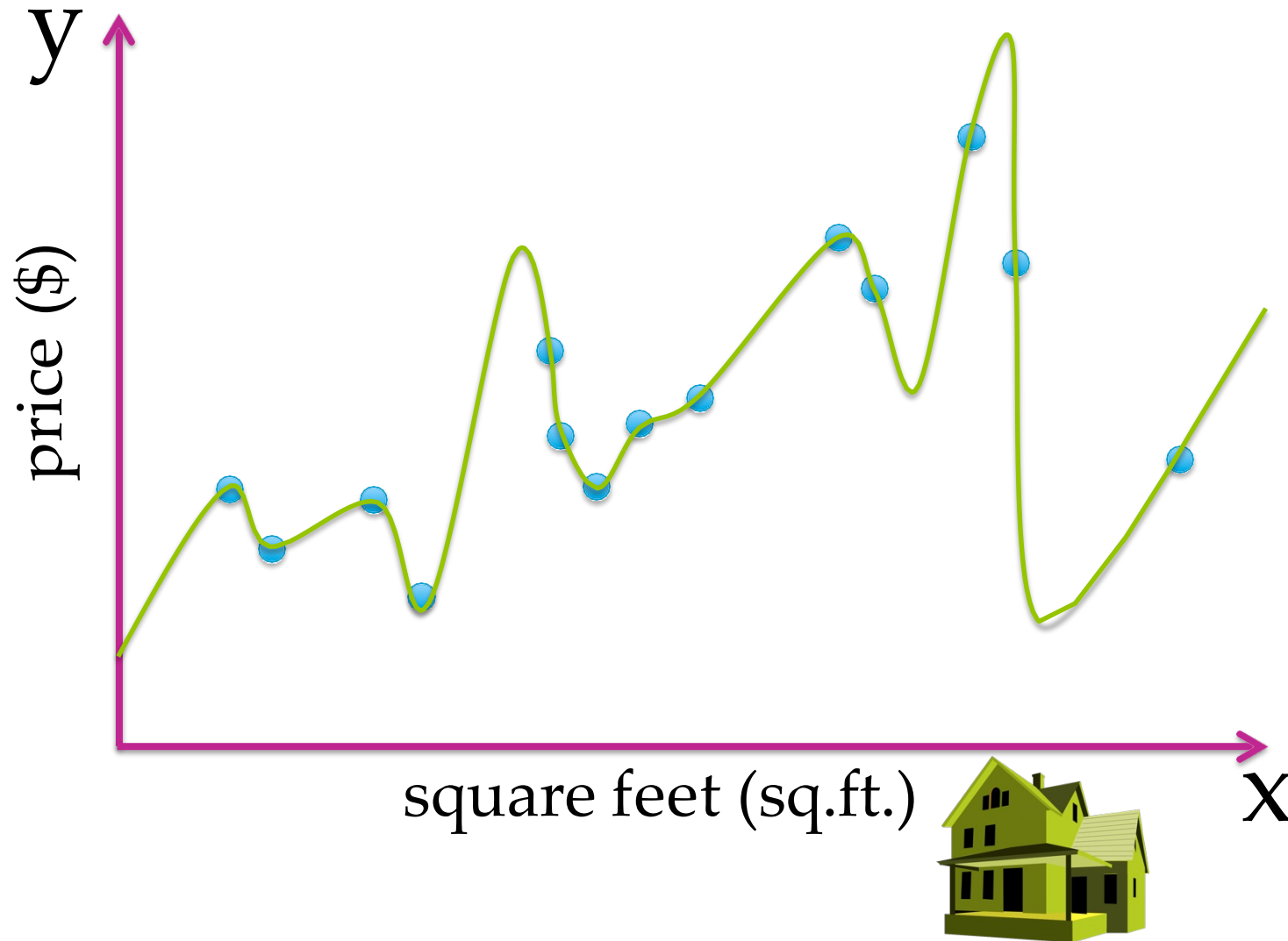
Predicting house prices

What about a quadratic function?



Predicting house prices

Even higher order polynomial

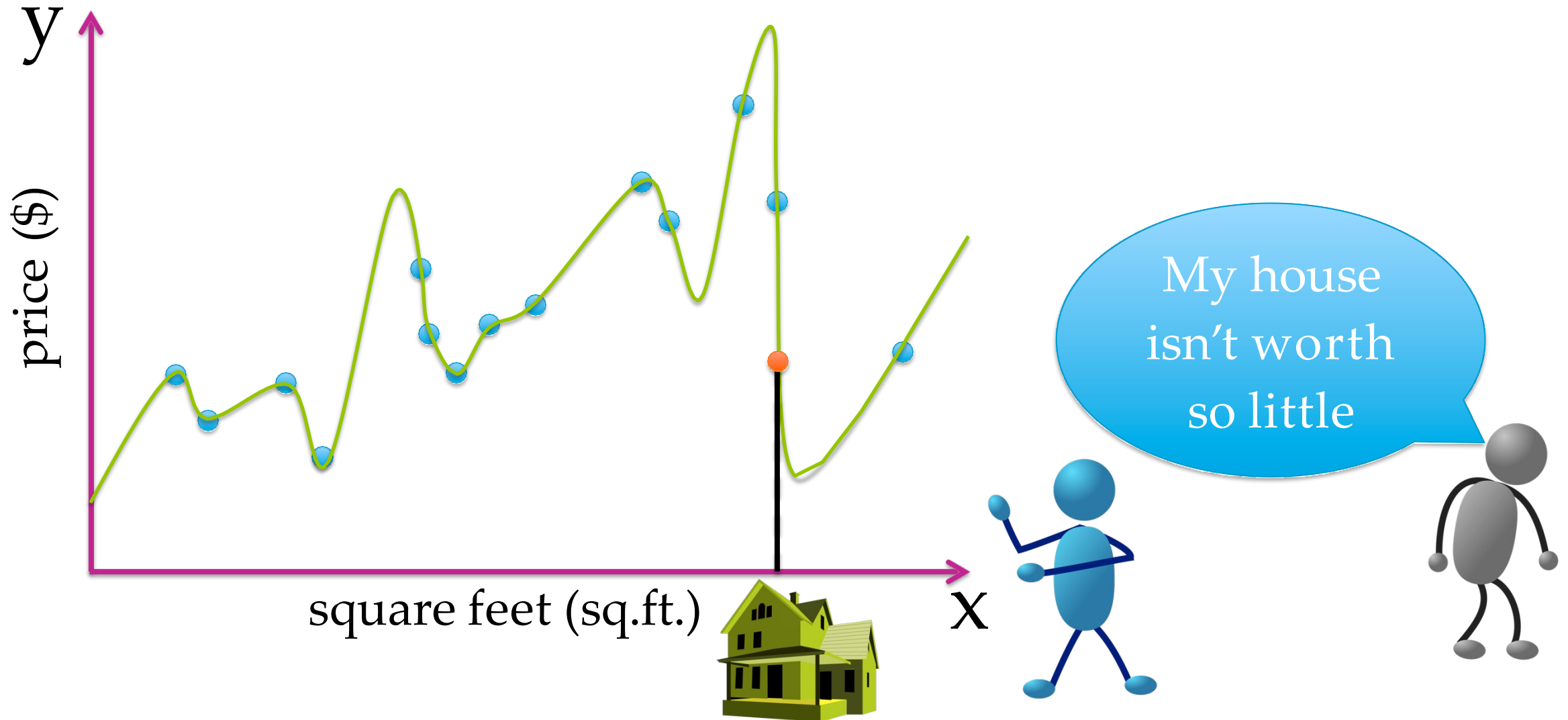


Still remember?
Residual sum of squares (RSS)

I can
minimize
your RSS

Predicting house prices

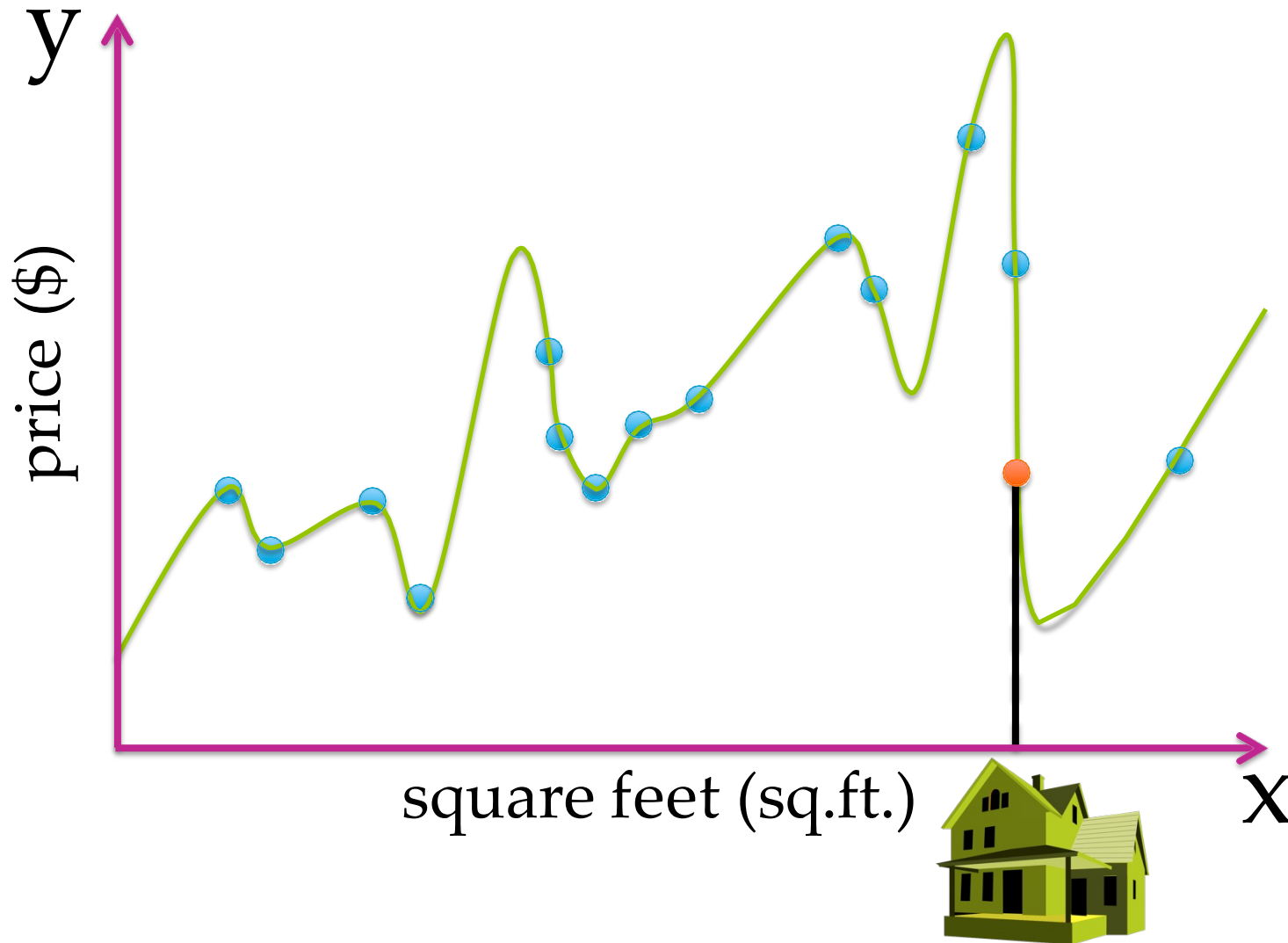
Do you believe this fit?



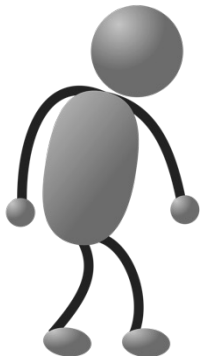
Evaluating

Predicting house prices

Do you believe this fit?

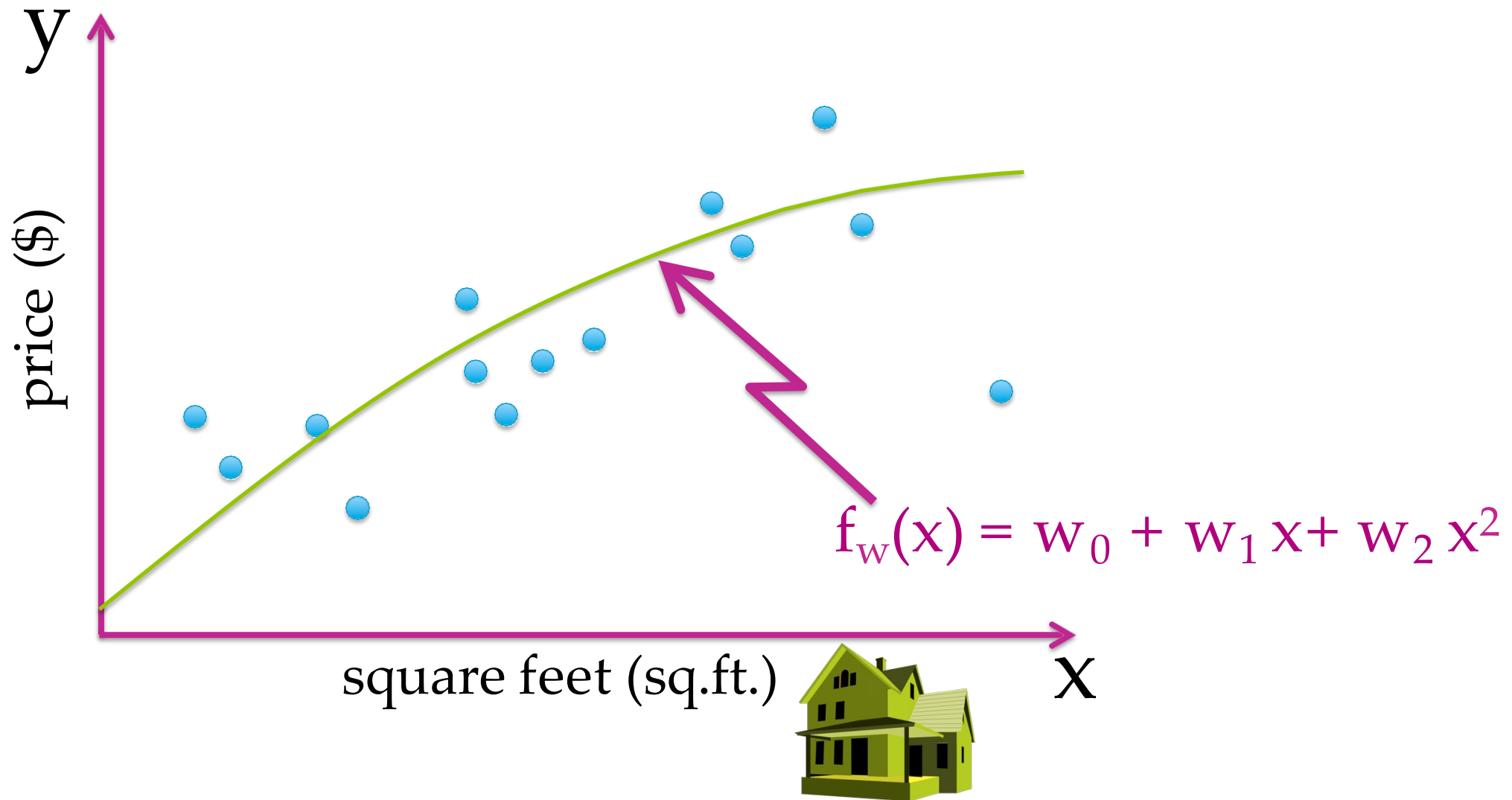


Minimizes RSS,
but **bad predictions**



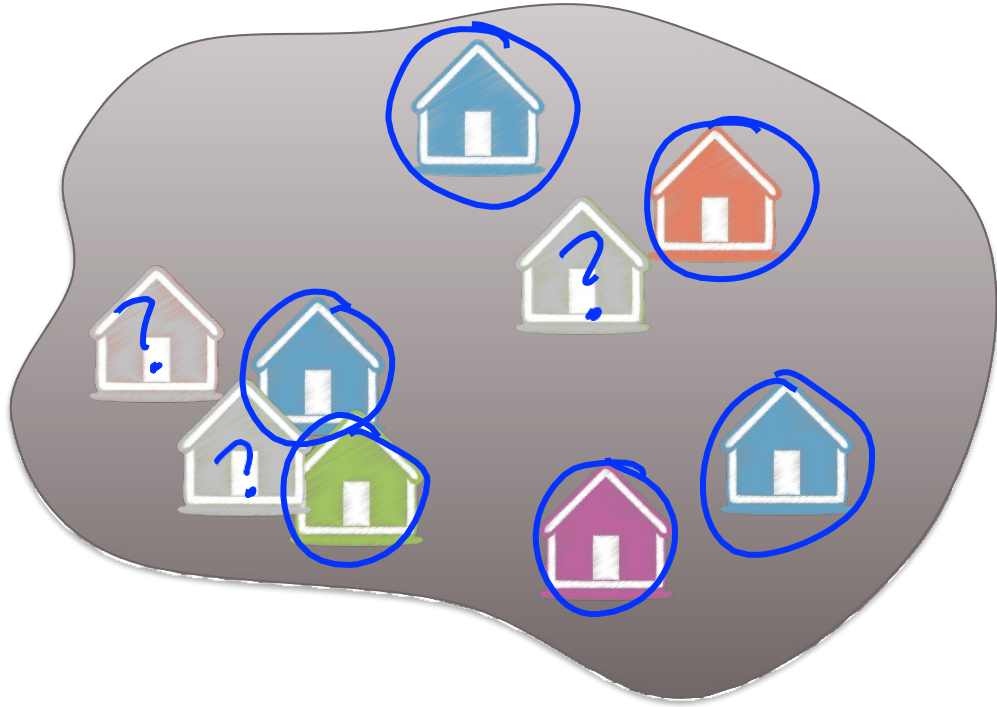
Predicting house prices

What about a quadratic function?



Predicting house prices

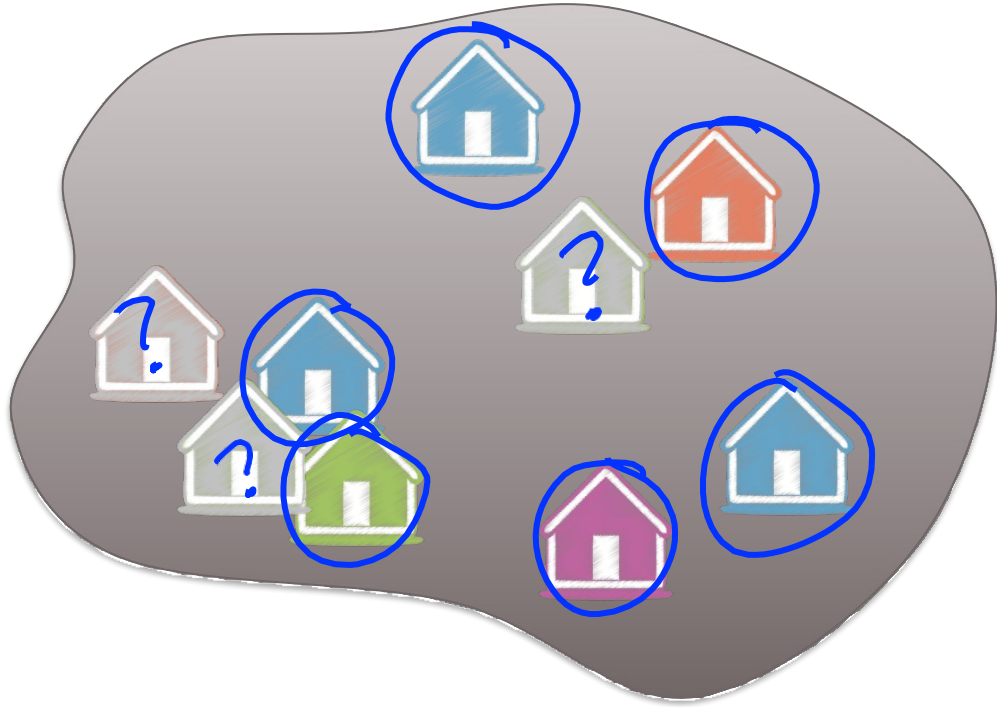
How to choose model order/complexity



- Want good predictions, but can't observe future
- Simulate predictions
 1. Remove some houses
 2. Fit model on remaining
 3. Predict heldout houses

Predicting house prices

Training/Test Split



- Want good predictions, but can't observe future
- Simulate predictions
 1. Remove some houses
 2. Fit model on remaining
 3. Predict heldout houses

Terminology: – training set

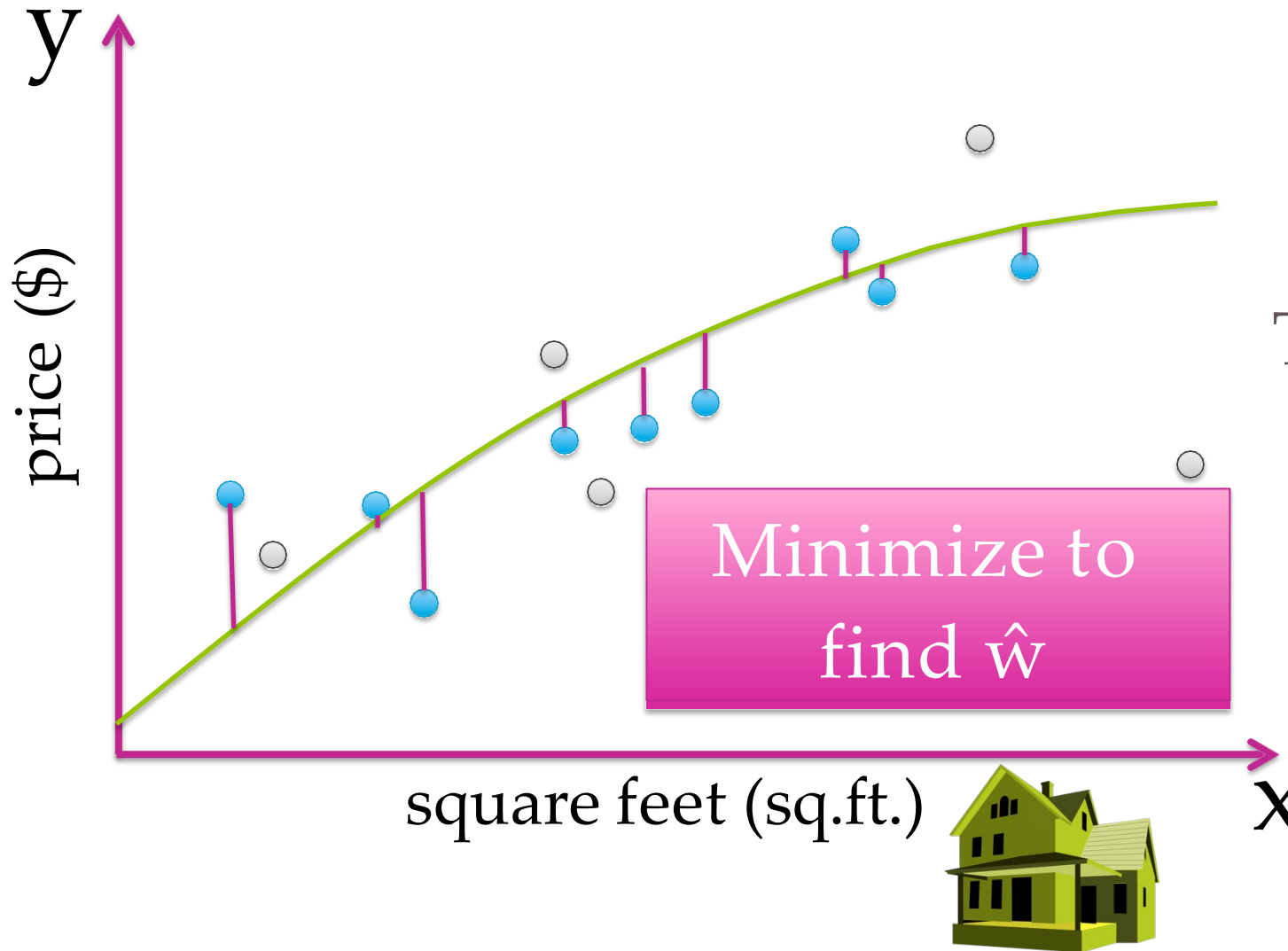


– test set



Predicting house prices

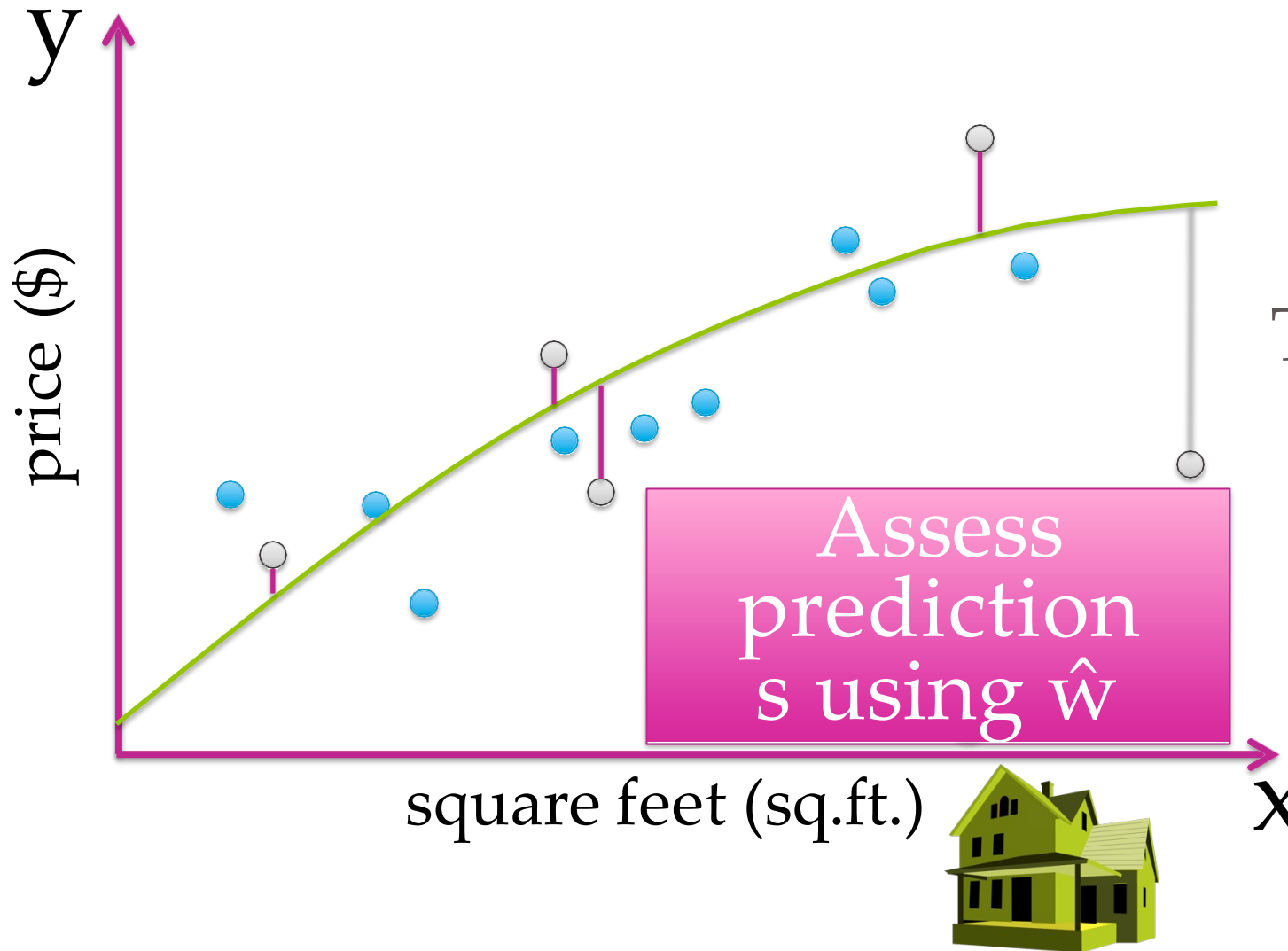
Training Error



$$\begin{aligned} \text{Training error } (w) = & (\$_{\text{train 1}} - f_w(\text{sq.ft.}_{\text{train 1}}))^2 \\ & + (\$_{\text{train 2}} - f_w(\text{sq.ft.}_{\text{train 2}}))^2 \\ & + (\$_{\text{train 3}} - f_w(\text{sq.ft.}_{\text{train 3}}))^2 \\ & + \dots [\text{include all} \\ & \quad \text{training houses}] \end{aligned}$$

Predicting house prices

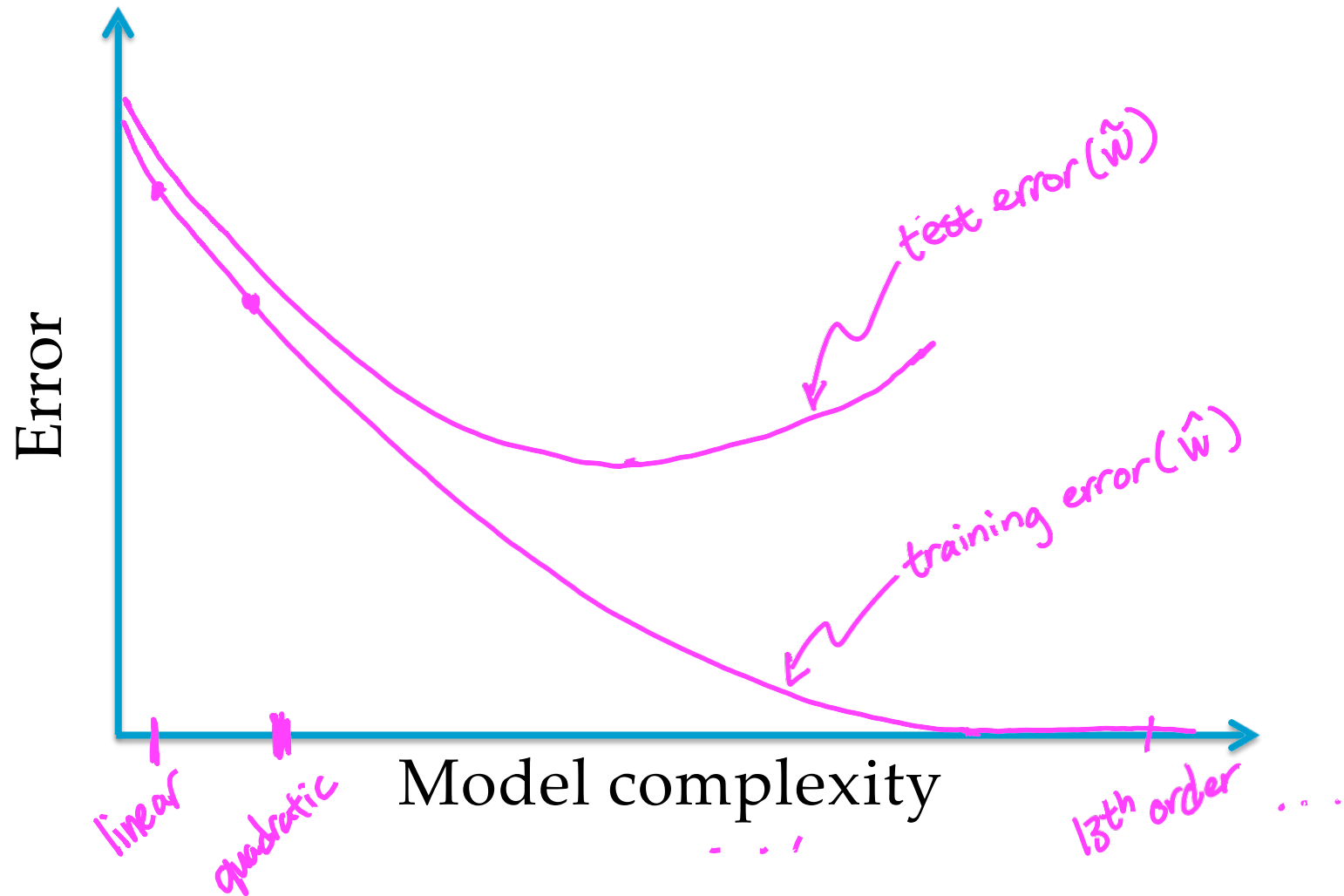
Test Error



$$\begin{aligned} \text{Test error } (\hat{w}) = & (\$_{\text{test 1}} - f_{\hat{w}}(\text{sq.ft.}_{\text{test 1}}))^2 \\ & + (\$_{\text{test 2}} - f_{\hat{w}}(\text{sq.ft.}_{\text{test 2}}))^2 \\ & + (\$_{\text{test 3}} - f_{\hat{w}}(\text{sq.ft.}_{\text{test 3}}))^2 \\ & + \dots [\text{include all test houses}] \end{aligned}$$

Predicting house prices

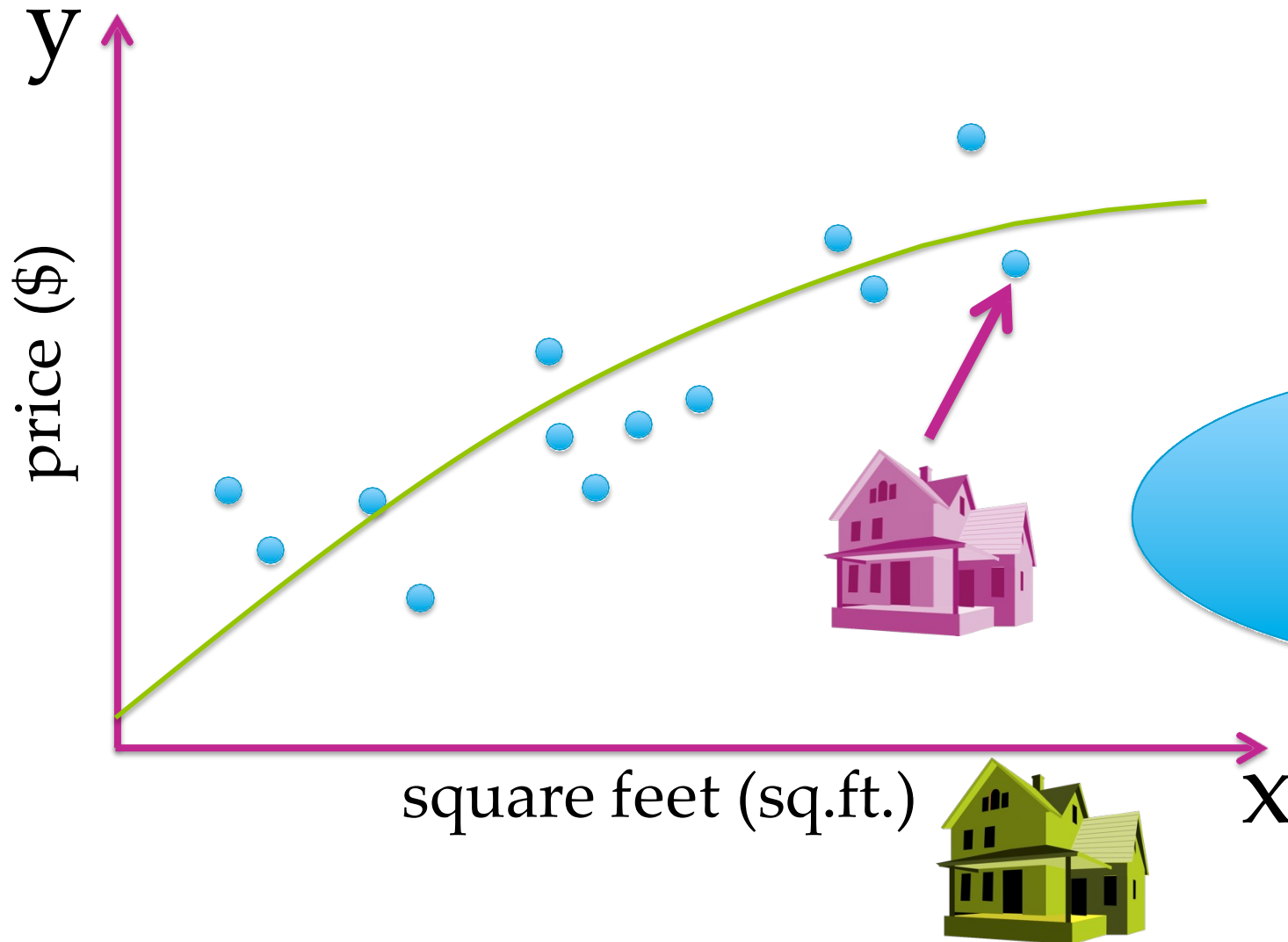
Training/Test Curves



Adding Other Features

Predicting house prices

Predictions just based on house size

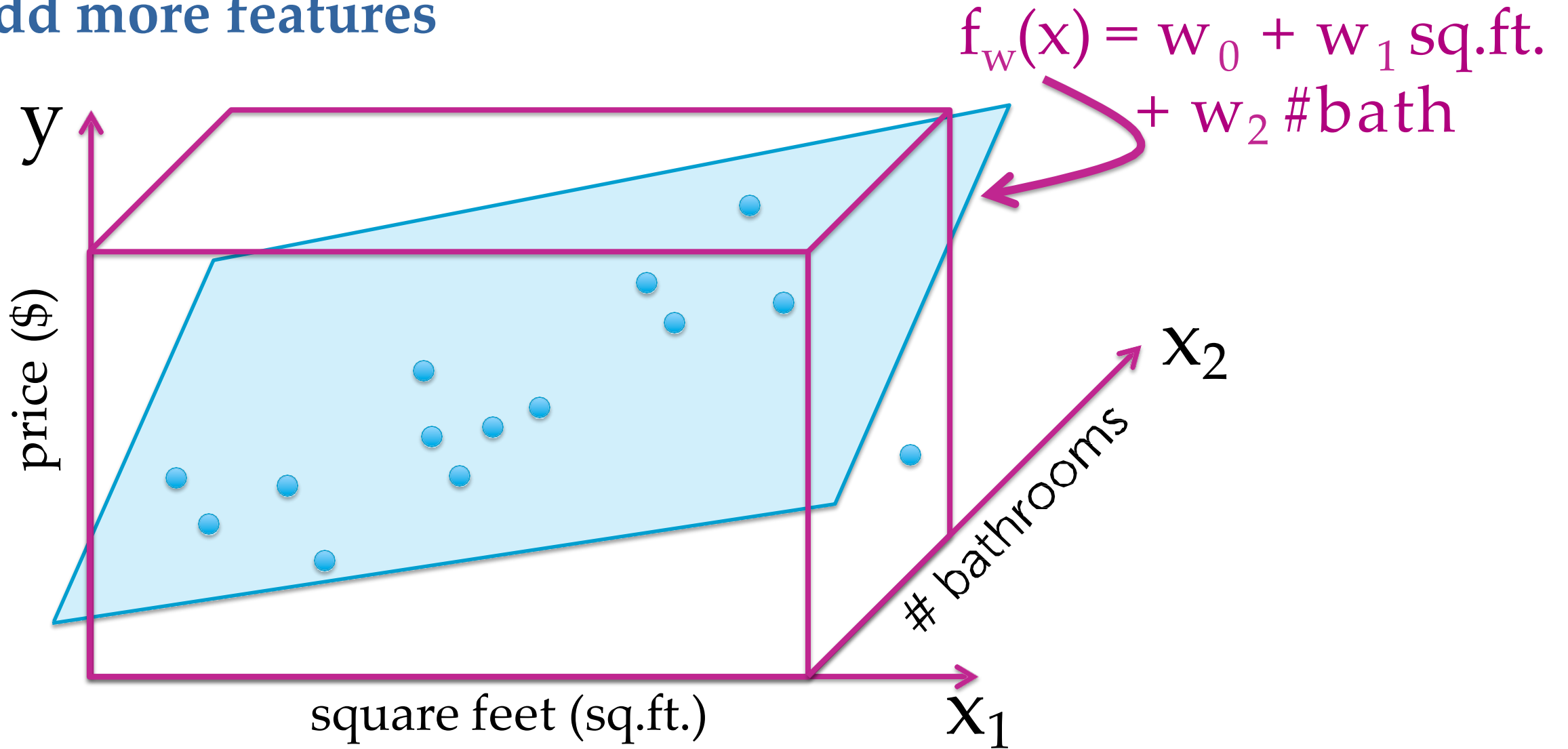


Only 1 bathroom!
Not same as my 3
bathrooms



Predicting house prices

Add more features



Predicting house prices

How many features to use?

- Possible choices:
 - Square feet
 - # bathrooms
 - # bedrooms
 - Lot size
 - Year built
 - ...

Other Regression Examples

Salary after learning this ML course



hard work



- How much will your salary be? ($y = \$\$$)
- Depends on x = performance in courses, quality of capstone project, # of forum responses, ...

Salary after learning this ML course



hard work



$$\hat{y} = \hat{w}_0 + \hat{w}_1 \text{ performance} + \hat{w}_2 \text{ capstone} + \hat{w}_3 \text{ forum}$$

informed by other students who
completed specialization

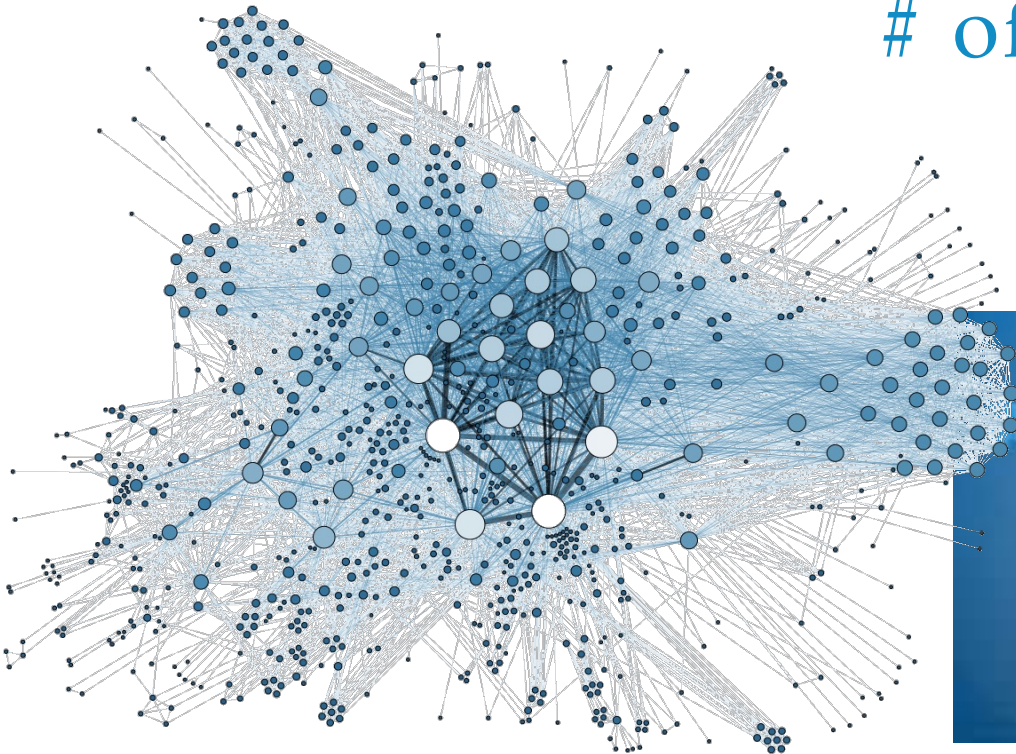
Stock prediction

- Predict the price of a stock
- Depends on
 - Recent history of stock price
 - News events
 - Related commodities



Tweet popularity

- How many people will retweet your tweet?
- Depends on # followers,
of followers of followers, features of
text tweeted, popularity of hashtag,
of past retweets,...



Smart houses

- Smart houses have many distributed sensors
- What's the temperature at your desk? (no sensor)
 - Learn spatial function to predict temp
- Also depends on
 - Thermostat setting
 - Blinds open/closed or window tint
 - Vents
 - Temperature outside
 - Time of day



Summary for Regression

What you can do now...

- Describe the input (features) and output (real-valued predictions) of a regression model
- Calculate a goodness-of-fit metric (e.g., RSS)
- Estimate model parameters by minimizing RSS
- Exploit the estimated model to form predictions
- Perform a training/test split of the data
- Analyze performance of various regression models in terms of test error
- Use test error to avoid overfitting when selecting amongst candidate models
- Describe a regression model using multiple features
- Describe other applications where regression is useful

Python Implementation

Python Implementation

Scikit-learn (Machine Learning Python Library)

- Scikit-learn is a powerful and widely used library for creating custom machine learning models. It is designed to be simple and efficient, making it accessible even to those who are not experts in machine learning.

Advantages of scikit-learn

- Easy to Use: Its clean and consistent API makes it beginner-friendly.
- Highly Flexible: scikit-learn provides support for a wide range of machine learning algorithms, from simple linear regression to complex ensemble methods.
- Integration with Data Formats: It can seamlessly handle datasets in formats like .csv (Comma Separated Values) using pandas or numpy for preprocessing.
- Extensive Documentation: scikit-learn comes with comprehensive documentation and examples to guide users through building and evaluating model

Install first if not installed: `$ pip install -U scikit-learn`

User guide: https://scikit-learn.org/stable/user_guide.html



Python Implementation

Load the data

```
import pandas as pd

# Load the data
data = pd.read_csv("data.csv")

# check the data
data.head()
```

id	date	price	bedrooms	bathrooms	sqft_living	
7129300520	20141013T000000	221900.0	3	1.0	1180	
6414100192	20141209T000000	538000.0	3	2.25	2570	
5631500400	20150225T000000	180000.0	2	1.0	770	
2487200875	20141209T000000	604000.0	4	3.0	1960	
1954400510	20150218T000000	510000.0	3	2.0	1680	
7237550310	20140512T000000	1225000.0	4	4.5	5420	
1321400060	20140627T000000	257500.0	3	2.25	1715	
2008000270	20150115T000000	291850.0	3	1.5	1060	
2414600126	20150415T000000	229500.0	3	1.0	1780	
3793500160	20150312T000000	323000.0	3	2.5	1890	

Python Implementation

Create a simple regression model

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
my_features = ['sqft_living']
train_data, test_data = train_test_split(data, test_size=0.2, random_state=0)
```

Set the features using in prediction
Split train/test sample

```
X_train = train_data[my_features]
y_train = train_data['price']
X_test = test_data[my_features]
y_test = test_data['price']
```

Split features and the target variable

```
sqft_model = LinearRegression()
sqft_model.fit(X_train, y_train)
y_pred = sqft_model.predict(X_test)
```

Create a regression model
using the features and
train sample

```
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
```

Evaluate the model

```
print(f"Mean Squared Error: {mse}")
print(f"R^2 Score: {r2}")
```

Python Implementation

Using the model to predict house price

```
house1 = data[data['id'] == '5309101200']
```

Suppose we want to predict the house
whose id is 5309101200

```
house1_features = house1[my_features]
```

```
house1_prediction = sqft_model.predict(house1_
```

```
print(house1_prediction) See the predicted price
```

```
>>[629976.0401234042]
```

```
print (house1['price'])
```

Check the actual price

```
>>[620000, ... ]
```



Tutorial

